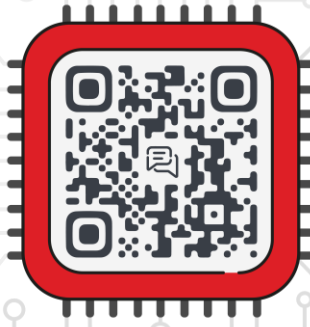




نظري

كتاب

64

7680 sp

كلية الهندسة المعلوماتية

السنة الثالثة

Programming Languages Extension

Programming
Languages



عمل طلابي

محتوى مجاني غير مخصص للبيع التجاري

RB Informatics;

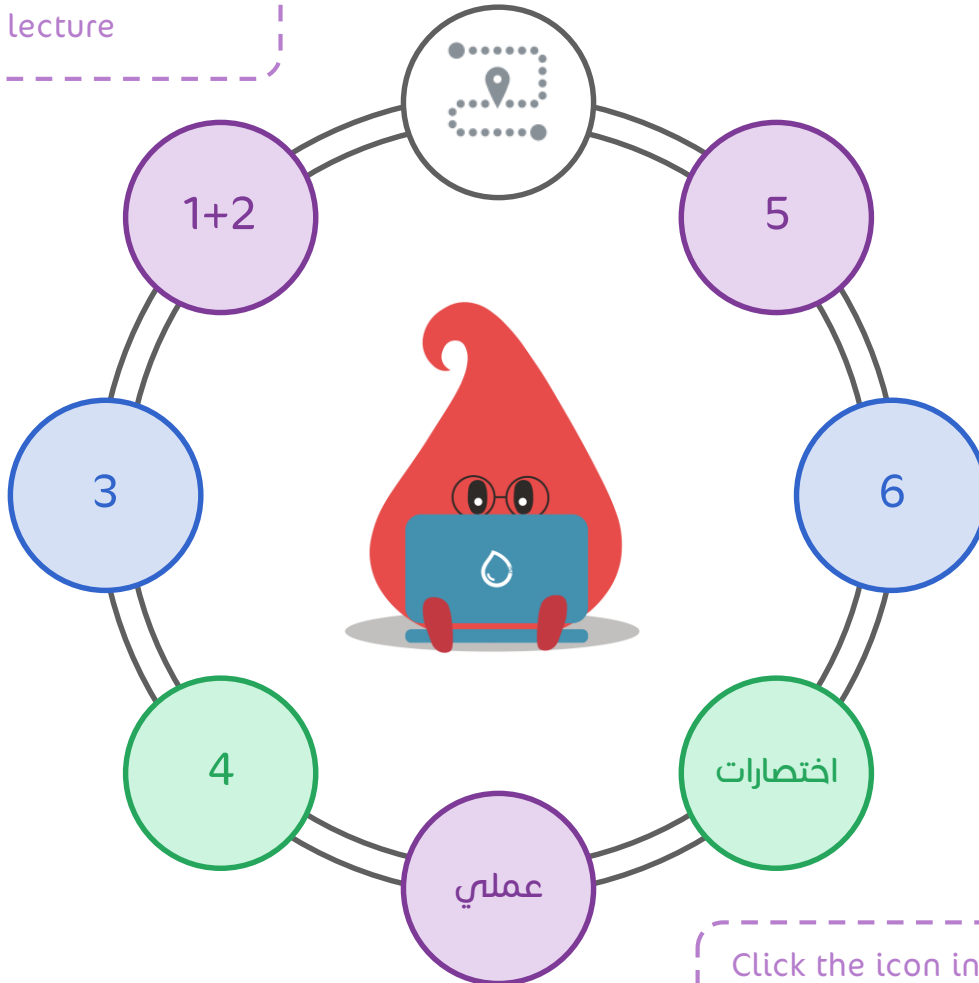
3/3/2025

لغات البرمجة

مرحباً يا أصدقاء ☺

نضع بين أيديكم هذا الملحق ونتمنى أن يكون عوناً لكم، وهو نموذج على نمط دورات المادة مستوحى من المحاضرات والدورات والمراجع، نؤوه على أنه لا يغني عن دراسة المحاضرات وأفكار المادة إنما لتثبيت معلوماتكم والتدرب على نموذج الدورات، مع تمنياتنا لكم بالتوفيق والنجاح.

Click on the number to go to
the lecture



Click the icon in the top left
to return to the first page



المحاضرة الأولى + الثانية

1) What does BRD stand for?

- | | |
|----------------------------------|------------------------------|
| A) Business Requirement Document | C) Budget Resource Document |
| B) Business Review Document | D) Business Risk Declaration |

A

2) What is the primary purpose of a BRD?

- | | |
|--|------------------------------------|
| A) To define the technical requirements of the project | C) To analyze financial risks |
| B) To clarify the project's goals and requirements | D) To prepare the project timeline |

B

3) Who is typically responsible for preparing a BRD?

- | | |
|--------------------|---------------------------|
| A) Marketing team | C) Business Analyst |
| B) Project manager | D) Technical support team |

C

4) Which of the following elements is NOT considered part of a BRD?

- | | |
|----------------------------|--------------------------------|
| A) Functional requirements | C) Estimated budget |
| B) Feasibility analysis | D) Non-functional requirements |

C

5) What is the primary role of a System Analyst?

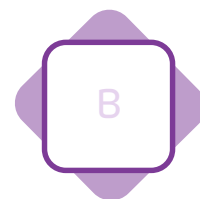
- | | |
|---------------------------|---|
| A) Design user interfaces | C) Identify and analyze system requirements |
| B) Write code | D) Test the system |

C



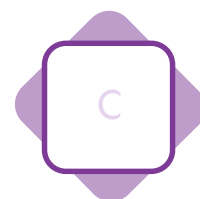
6) What is the purpose of Use Cases?

A) Document source code	C) Write internal system design
B) Describe how users interact with the system	D) Improve system performance



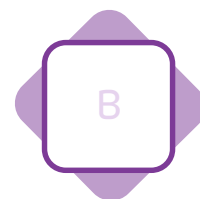
7) What is the main content of the System Requirements Specification (SRS)?

A) Source code	C) Functional and non-functional requirements
B) Database design	D) Team member list



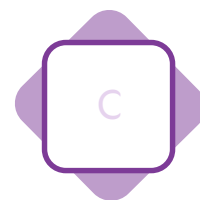
8) What is the role of the UI/UX team?

A) Collect requirements	C) Write code
B) Design user interfaces and user experience	D) Document Use Cases



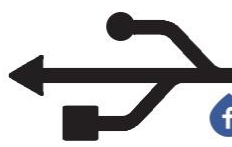
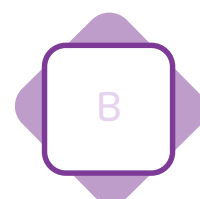
9) Which of the following is a functional requirement in the login process?

A) Design a colorful interface	C) Display an error message when incorrect data is entered
B) Write source code	D) Improve system performance



10) What is the System Requirements Specification (SRS) document?

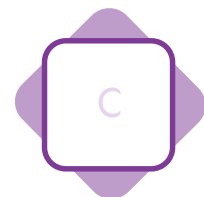
A) A document that explains the system's technical design details	C) A document with the project timeline
B) A document containing all functional and non-functional system requirements	D) A document explaining how to test the system





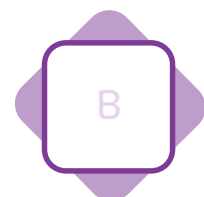
11) Which of the following is not usually included in the SRS document?

A) System description	C) Project budget
B) Performance requirements	D) Security requirements



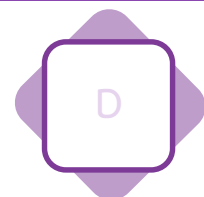
12) One of the main benefits of the SRS document?

A) Reducing development time	C) Increasing project costs
B) Improving software quality by clarifying requirements	D) Reducing the number of meetings with stakeholders



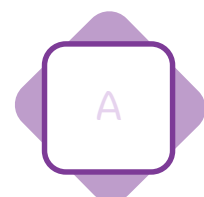
13) Which department receives the SRS file and the final screen design?

A) Marketing Department	C) Human Resources Department
B) Sales Department	D) Development Department



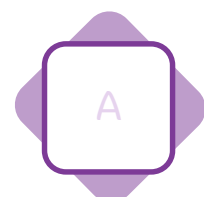
14) What is the main role of the Front-end department

A) Writing screen code based on the design	C) Analyzing requirements
B) Programming system logic	D) Preparing project reports



15) What is the main role of the Back-end department?

A) Programming the logic that makes the screens functional	C) Designing interfaces
B) Improving user experience	D) Preparing the SRS file



16) In which year was the WWW (World Wide Web) invented?

A) 1975

C) 1960

B) 1995

D) 1989

D

17) What was the main reason behind the invention of the WWW??

A) Creating electronic games

C) Sending text messages

B) Sharing information between connected devices

D) Data storage only

B

18) Which language is used to describe the structure and layout of a web page?

A) CSS

C) HTML

B) JavaScript

D) PHP

C

19) Which language is used to style colors and fonts on web pages?

A) CSS

C) HTML

B) JavaScript

D) Python

A

20) Which of the following languages is used to add interactivity to web pages?

A) CSS

C) HTML

B) JavaScript

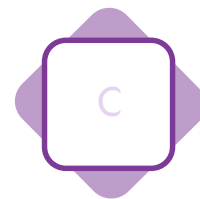
D) SQL

B



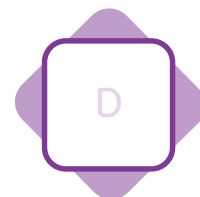
21) What is the main difference between a Website and a Web Application?

A) A Website contains only images	C) A Website is static, while a Web Application allows interaction with data
B) A Web Application uses only HTML	D) There is no difference between them



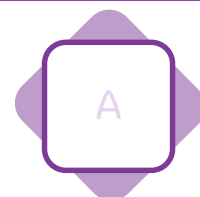
22) What tool is used to store data in interactive applications?

A) API	C) HTML
B) CSS	D) Database



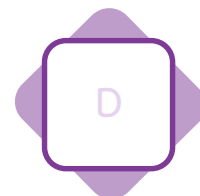
23) What is the architecture model where the interface, logic, and data are combined in a single layer without the need for the internet or servers?

A) 1-Tier Architecture	C) N-Tier Architecture
B) 2-Tier Architecture	D) Client-Server Model



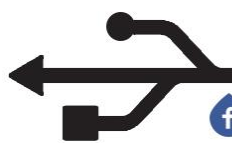
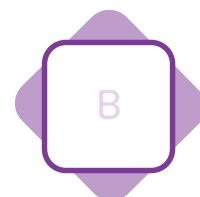
24) Which of the following is a disadvantage of 1-Tier Architecture?

A) Lack of communication between devices	C) Inability to transfer data easily
B) Limited performance depending on device capabilities	D) All of the above



25) How is the application divided in 2-Tier Architecture?

A) Into three layers: Interface, Logic, Data	C) Into Backend and Database
B) Into two parts: Client and Server	D) Into five different layers





26) What is the main issue in 2-Tier Architecture that led to the development of N-Tier Architecture?

A) Difficulty in updating code on all devices when errors occur

C) Weak database performance

B) Inability to connect to the internet

D) Lack of multimedia support

A

27) What concept emerged in 2-Tier Architecture due to logic concentration in the Client?

A) Thin Client

C) Cloud Computing

B) Fat Client

D) Virtualization

B

28) What is the role of the GUI layer in N-Tier Architecture?

A) Data processing

C) Displaying graphical interfaces without logic

B) Performing calculations

D) Storing data

C

29) Which layer is responsible for processing APIs in N-Tier Architecture?

A) GUI

C) Data

B) Logic

D) Server

B

30) In N-Tier Architecture, which layer holds the data without performing any processing operations?

A) GUI

C) Data

B) Logic

D) Server

C

31) What is the primary goal of N-Tier Architecture?

A) Improving data security and load distribution

C) Merging the interface and logic together

B) Reducing internet dependency

D) Using only low-performance devices

A

32) Which of the following is an example of 1-Tier Architecture?

A) An independent desktop application without server connection

C) ERP system distributed across multiple servers

B) Web application depending on a server-side database

D) Mobile app relying on API to communicate with the server

A

33) Which of the following is a characteristic of 2-Tier architecture ?

A) Presentation and business logic are separated into different layers

C) It provides better scalability compared to N-Tier architecture

B) The client interacts directly with the database

D) None of the above

B

34) who is responsible for creating the Mockup?

A) Business Analyst

C) System Analyst

B) UI/UX Designer

D) Project Manager

B

35) Which option best describes the concept of a "Fat Client"?

A) Fully relies on the server for data processing

C) Only stores data without performing any processing.

B) Processes a large part of data and operations locally instead of depending on the server.

D) Works only in weak network environments.

B



المحاضرة الثالثة

1) Which of the following is not one of the essential components of the web?

- | | |
|---------|---------|
| A) HTML | C) HTTP |
| B) URL | D) Port |

D

2) HTML is the primary markup language used for designing and structuring web pages

- | | |
|----------|------|
| A) True | C) ♥ |
| B) False | D) ♥ |

A

3) Which of the following represents the correct format for a URL?

- | | |
|-------------------------------------|--------------------------|
| A) http://www.google.com/index.html | C) http://www.google.com |
| B) FTP://www.google.com:8090 | D) a & b & c are True |

D

4) In the address https://www.google.com/home.html, what does home.html represent

- | | |
|----------------|--------------|
| A) Page name | C) port name |
| B) Domain name | D) Host name |

A

5) Which of the following is a fundamental component of the URL structure

- | | |
|----------------|-----------------|
| A) Port Number | C) Path |
| B) Domain name | D) Query String |

B



6) The Anchor is a non-essential component of the URL, typically used for:

A) Multi page Design

C) ♥

B) One Page Design

D) ♥

B

7) What is the function of DNS?

A) Storing website content

C) Securing internet connections

B) Translating domain names to IP addresses

D) Managing email servers

B

8) The default port values for the HTTP and HTTPS protocols respectively are:

A) 80 - 443

C) 80 - 22

B) 443 - 80

D) 80 - 23

A

ملاحظة: إن 23 هي قيمة افتراضية لبروتوكول telnet، و22 قيمة افتراضية لبروتوكول SSH

9) What is the default value for the FTP protocol:

A) 143

C) 21

B) 23

D) 25

C

10) Which is the correct format for passing parameters within the URL:

A) ?id=31&name=Omar

C) /id=31&name=Omar

B) /id=23_name=omar

D) None of the above

A

11) communication between the client and the server is done through Request and Response

A) True

C) ♥

B) False

D) ♥

A

12) What is one of the main benefits of using an HTTP Proxy between the server and client?

A) Increase download speed

C) Enhance network communication security

B) Protect user data privacy

D) None of the above

B

13) What do Response bodies typically consist of?

A) Different data types such as text, HTML, XML, JSON, images, or files

C) Protocol version

B) Status codes

D) A & B are true

A

14) One of the following sentences is correct:

A) The GET method contains a body and is used to display specific data from the server

C) The POST method and GET method perform the same tasks

B) The POST method contains a body and is used to modify previously stored data

D) The POST method contains a body and is used to add data to the server

D

15) Which of the following statements is true regarding the security difference between POST and GET requests:

A) POST is considered more secure than GET because GET sends data as part of the URL address while POST passes data within the Request body, making it harder to intercept

C) The length of the URL in GET requests is limited, while there are no restrictions on URL length in POST requests

B) Data is sent as part of the URL address in both POST and GET methods

D) Both A and B are true.

A

ملاحظة: لو لم يذكر في السؤال regarding the security difference between POST and GET requests كان الجواب D

16) What is the main difference between the GET method and HEAD method

A) The GET method is faster and returns the response without a body

C) The HEAD method is faster and returns the response without a body

B) The GET method is faster and returns the response with a body

D) The HEAD method is faster and returns the response with a body

C

17) Which HTTP request method is used by a client to update an existing resource on the server?

A) Edit

C) Update

B) put

D) modify

B

18) What is the main difference between the GET method and HEAD method?

A) The GET method is faster and returns the response without a body

C) The HEAD method is faster and returns the response without a body

B) The GET method is faster and returns the response with a body

D) The HEAD method is faster and returns the response with a body

C



19) Which HTTP request method is used by a client to delete an existing resource on the server?

A) Remove

C) ♥

B) Delete

D) ♥

B

12) Which HTTP request method is used by a client to submit data to be processed to the server?

A) get

C) post

B) put

D) delete

C

21) Which HTTP request method is used by a client to request data from a server?

A) get

C) post

B) put

D) delete

A

23) which http status code indicates that the requested resource has been permanently moved to a new location?

A) 400

C) 301

B) 200

D) 500

C

24) which http status code indicates that there is a syntax error?

A) 400

C) 301

B) 200

D) 500

A



25) Choose the correct answer:

A) The DNS translates the Domain Name to an IP	C) Tasks within a single device with the same IP are distinguished by the port
B) The DNS translates the IP to a Domain Name	D) A & B



26) Choose the correct answer:

A) The HEAD method is similar to GET but differs in the response	C) POST cannot perform the tasks of PUT and DELETE
B) POST can always be replaced by GET, but the opposite is not true	D) All of the above are incorrect



27) Status code 400 indicates:

A) Bad request	C) Unauthorized
B) Not found	D) Service unavailable

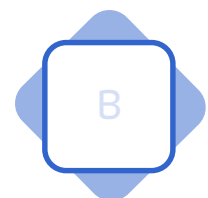


28) Choose the correct output in HTML:

```

!< DOCTYPE html >
< html >
  < head >
    < title>Page Title</title >
  /< head >
  < body >
    < u><i>Hello</i></u >
  /< body >
/< html >
    
```

A) <i>Hello</i>	C) Hello
B) <u>Hello</u>	D) Hello





29) PHP language: (Choose the wrong answer)

A) Case-sensitive

C) It can be printed using echo or print

B) Can be used within HTML as follows
<?....>

D) all of the above is correct

A

30) `route::get('/exam', examController @ examTest)`

A) Handling exam test within the function

C) Handling exam within the function examTest

B) Receiving exam in examController

D) B & C

D

31) Which HTTP status code indicates that the client's request was successful but there is no content to return?

A) 200

C) 400

B) 204

D) 500

B

32) Which protocol is used for secure communication over a computer network?

A) HTTP

C) SSH

B) FTP

D) Telnet

C

33) Which HTML attribute is used to specify an inline CSS style for an element?

A) Class

C) style

B) id

D) CSS

C

34) Which HTML element is used to create a hyperlink?

- | | |
|-----------|----------------|
| A) <link> | C) <href> |
| B) <a> | D) <hyperlink> |

B

35) Which HTTP status code indicates that the client's request cannot be completed due to invalid syntax?

- | | |
|--------|--------|
| A) 200 | C) 404 |
| B) 400 | D) 500 |

B

36) Which HTML element is used to define a table cell?

- | | |
|---------|------------|
| A) <td> | C) <tr> |
| B) <th> | D) <table> |

A

37) Command to create a new Laravel Project?

- | | |
|-----------------------|----------------------|
| A) Php artisan new | C) Php artisan make |
| B) Php artisan create | D) None of the above |

D

38) Laravel uses which template engine by default?

- | | |
|----------|-------------|
| A) Twig | C) Mustache |
| B) Blade | D) smarty |

B

39) Which folder in Laravel is used to store migration files?

- | | |
|--------------------------|-------------------------|
| A) App/migrations | C) Database/ migrations |
| B) Resources/ migrations | D) Config/ migrations |

C



40) Which HTTP request method is used to retrieve data from a server?

| | |
|---------|-----------|
| A) Get | C) Put |
| B) Post | D) Delete |



41) HTTP proxy is:

| | |
|----------------------|---|
| A) Internet provider | C) It is possible to connect between the server and the client without using an HTTP proxy. |
| B) Server | D) A & B |



42) Choose the correct answer

| | |
|--|-------------------------------|
| A) The URL can only contain the protocol HTTP or HTTPS | C) Both A and B are incorrect |
| B) It is not possible to navigate within parts of the page through the URL | D) Both A and B are correct |



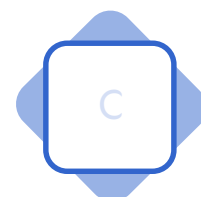
43) Which of the following is not part of performance testing?

| | |
|--------------------------------------|-----------------|
| A) Response time for client requests | D) Memory usage |
| B) Request time | E) Load test |
| C) Stress test | F) B & C |



44) Choose the correct answer

| | |
|---|---|
| A) Status codes starting with 1 indicate server errors. | C) Status codes starting with 3 request additional action from the user other than the sent request (indicate redirection responses). |
| B) Status codes starting with 2 indicate bad responses. | D) Status codes starting with 4 indicate server errors |



45) One of these options is not part of the response

| | |
|--------------------|------------------|
| A) Method | C) Response body |
| B) Version of http | D) Status code |



أشهر البروتوكولات المستخدمة:

| Protocol name | Protocol use | Port number |
|---------------|---|-------------|
| HTTP | Document Transfer | 80 |
| FTP | File Transfer | 21 |
| SSH | Secure Server Access | 22 |
| SMTP | Simple Mail Transfer Protocol | 25 |
| HTTPS | Hypertext Transfer Protocol Secure | 443 |
| Telnet | Remote Communication Protocol, which is not secure because it does not include encryption | 23 |
| POP3 | Post Office Protocol | 110 |
| IMAP | Internet Message Access Protocol | 143 |



المحاضرة الرابعة

1) HTTP protocol is a stateless protocol.

A) True

C)

B) False

D)

A

2) User state and related information are stored only through cookies

A) True

C)

B) False

D)

B

3) Which type of storage is used if the language settings and display mode (Light Mode/Dark Mode) are changed, the user leaves the website, and upon revisiting, the previously applied modifications remain unchanged?

A) Session Storage

C) Cookies

B) Database Storage

D) None of the above

C

4) Choose the correct answer that explains the key differences between sessions and cookies:

A) The session is stored on the server and takes up a large size, while cookies are stored on the device and are used to store simple data.

C) The session persists after closing the browser or logging out, while the lifetime of cookies is determined by the user.

B) The session is stored on the server, while cookies are stored on the device, and both have a high level of security.

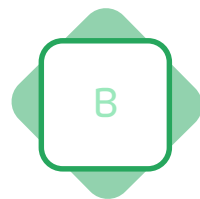
D) All of the above are correct

A



5) commerce websites always use Session Storage to store user data

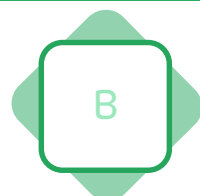
| | |
|----------|------|
| A) True | C) ♥ |
| B) False | D) ♥ |



ملاحظة: تستخدم ال session في مواقع التجارة الالكترونية لحفظ بيانات مؤقتة مثل حالة سلة التسوق قبل إتمام الشراء والبيانات التي يحتاجها المستخدم أثناء الجلسة، أما لحفظ بيانات تسجيل الدخول لفترة طويلة (حسب مدة صلاحيتها) كاسم المستخدم وكلمة المرور تحفظ في ال cookies، بينما يتم حفظ البيانات الدائمة كمعلومات الحساب، الطلبات، وسجلات الدفع ضمن ال database storage.

6) Session Storage is a highly secure method for storing sensitive data

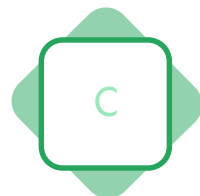
| | |
|----------|------|
| A) True | C) ♥ |
| B) False | D) ♥ |



ملاحظة: ال Session أكثر أماناً وحماية من ال cookies لكنها ليست آمنة بالكامل فهي غير مشفرة ويمكن الوصول إليها.

7) How does a web server typically identify a user's session?

| | |
|-------------------------------|--|
| A) By the user's IP address | C) By using a unique session identifier (session ID) |
| B) By the user's browser type | D) By storing the user's data in a cookie |



8) Which of the following is Not a characteristic of cookies?

| | |
|--|--|
| A) Cookies are stored on the client-side | C) Cookies can only store data in string format. |
| B) Cookies have an expiration time | D) Cookies can store unlimited amounts of data. |





9) Why did the concept of Software Architecture Patterns emerge?

| | |
|--|--|
| A) to manage software complexity, improve maintainability, facilitate collaboration, and enhance reusability | C) To make the code more complex |
| B) To reduce the cost of development | D) To speed up the development process without regard to quality |



10) In the MVVM pattern, how does the ViewModel communicate with the view?

| | |
|-------------------------------------|----------------------------|
| A) Through direct method calls | C) Through event listeners |
| B) Through data binding or commands | D) Through HTTP requests |



11) The correct answer is:

| | |
|---|---|
| A) MVC: The model connects directly to the view | C) MVVM: The view-model can specify the view to request a service |
| B) MVP: The model connects to the view | D) All of the above are correct |

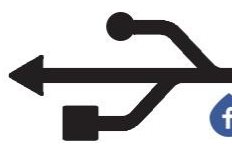


12) The correct answer is:

| | |
|---|---|
| A) MVP is the initial application that contains the essential features required by the customer | C) UX makes the application more user-friendly. |
| B) UI makes the application more functional | D) A & C are true |



1. UX makes the application more functional / UI makes the application more user-friendly.





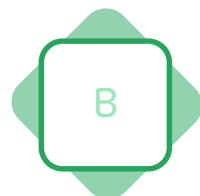
13) In Laravel, the function of middleware is:

| | |
|---|--|
| A) to intercept requests before they reach the controller | C) A middleware verifies the user's identity before accessing the application's services |
| B) to intercept requests after they reach the controller | D) None of the above. |



14) The role of model in MVVM pattern:

| | |
|---|---|
| A) Represents user interface components. | C) manage communication between ViewModel and view. |
| B) it's encapsulates the application's business logic and data. | D) handle user input events. |



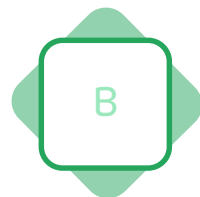
15) To create a controller in Laravel:

| | |
|--|--|
| A) php artisan make:controller controllename | C) php artisan make:controllename controller.. |
| B) php artisan create:controller controllename | D) php artisan creat:controllename controller. |



16) Command to get the route list in Laravel:

| | |
|---------------------------|-------------------------|
| A) Php artisan list | C) Php artisan route |
| B) Php artisan route:list | D) Php artisan list:all |





17) Which directory contains the application's models controllers and views in Laravel?

| | |
|-----------|-----------|
| A) app | C) public |
| B) events | D) http |

A

18) Laravel is based on which design pattern?

| | |
|--------|----------------------|
| A) MVC | C) MVVM |
| B) MVP | D) None of the above |

A

19) What is the primary purpose of cookies on web application?

| | |
|----------------------------|---------------------|
| A) Storing user preference | C) Managing session |
| B) Authenticating user | D) All of the above |

D

20) To connect the ViewModel with the View in MVVM:

| | |
|------------------------------------|---------------------------|
| A) Through direct method | C) Through event listener |
| B) Through binding data or command | D) None of the above |

B

21) Cookies are stored on the:

| | |
|----------------|----------------------|
| A) Client side | C) A & B are true |
| B) Server side | D) None of the above |

A



22) The correct answer is:

| | |
|--|--|
| A) Only MVC can specify the view to request a service | C) MVVM can't specify the view to request a service. |
| B) Both MVC & MVP can specify the view to request a service. | D) B & C are true. |

D

23) What is the role of Live Data in the MVVM architecture?

| | |
|---|--|
| A) Store data only without making it available to any other view. | C) Store data requested by view and allow other views to access it |
| B) Store data and return it directly to the view that requested it. | D) Manage computational processes inside the view. |

C

24) Which of the following statements are true about cookies?

| | |
|--|---|
| A) Cookies are sent with every HTTP request to the server | C) The expiration period of cookies can be defined by the developer |
| B) Users can modify or delete cookies through browser settings | D) All of the above |

D

25) Which of the following options is not a characteristic of cookies?

| | |
|---|---|
| A) Can be stored on the user's device for a long period | C) Can contain identifying data like user preferences |
| B) Can store very large data such as image files | D) Can control their expiration period upon creation |

B



المحاضرة الخامسة

1) What is one of the advantages of Web Services?

A) They require writing new code from scratch for each service request.

C) They only work with a specific programming language.

B) They allow reusing published code on the internet without needing to rewrite it.

D) They can only be used in local applications.

B

2) How can the Google Translate API be used in application development?

A) The developer must write the translation code entirely from scratch.

C) The translation service can only be used through Google's official website.

B) The translation service can be called online without the need to write new code.

D) The translation service requires downloading special software on the local device.

B

3) What is the service that enables text translation using an API?

A) Google Drive

C) Google Maps

B) Google Translate

D) Google Photos

B

4) What language is used to describe SOAP services?

A) JSON

C) HTML

B) XML

D) WSDL

D

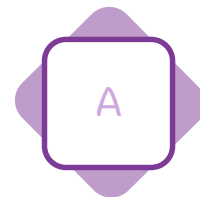
ملاحظة:

WSDL هي لغة مبنية على XML لكنها مخصصة لوصف خدمات ال SOAP، أما ال XML لغة عامة لنقل البيانات.



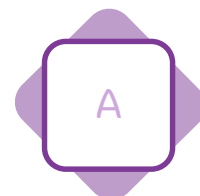
5) What is the difference between Authentication and Authorization?

| | |
|---|--|
| A) Authentication is about verifying the identity, whereas Authorization is about granting permissions. | C) There is no difference between them |
| B) Authorization is verifying the identity, while authentication is granting permissions. | D) none of the above |



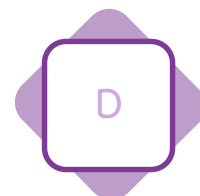
6) Which of the following methods are considered unsafe in the Authentication process?

| | |
|-------------------------|--------------|
| A) Basic Authentication | C) API Key |
| B) OAuth 2.0 | D) Biometric |



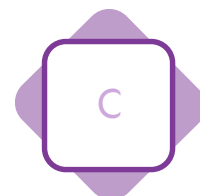
7) What is the most commonly used method in the Authentication process?

| | |
|-------------------------|--------------|
| A) Basic Authentication | C) API Key |
| B) Bearer Token | D) OAuth 2.0 |



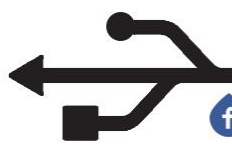
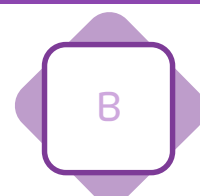
8) What is the Token used to request a new Access Token when it expires?

| | |
|-----------------|------------------|
| A) Access Token | C) Refresh Token |
| B) API Key | D) Bearer Token |



9) which method relies on something the user possesses in the Authentication process?

| | |
|-----------------------|--------------|
| A) Something you know | C) Biometric |
| B) Something you have | D) API Key |



10) which method relies on the biometric characteristics of the user in the Authentication process?

- | | |
|-----------------------|--------------|
| A) Something you know | C) Biometric |
| B) Something you have | D) API Key |

C

11) which method relies on the user's knowledge of specific information such as a password in the Authentication process?

- | | |
|-----------------------|--------------|
| A) Something you know | C) Biometric |
| B) Something you have | D) API Key |

A

12) What is the relative duration between the Access Token and the Refresh Token?

- | | |
|---|--|
| A) The Access Token has a longer duration than the Refresh Token. | C) both tokens have the same duration. |
| B) The Refresh Token has a longer duration than the Access Token. | D) none of the above. |

B

13) which authentication mechanism involves issuing tokens that expire after a certain period of time?

- | | |
|--------------------------|-----------------------|
| A) Basic Authentication. | C) Both a and b. |
| B) OAuth2. | D) none of the above. |

B

14) SOAP can be described:

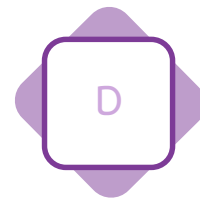
- | | |
|--|----------------------------|
| A) Using a WSDL file. | C) Both a and b are true. |
| B) It can be described using XML only. | D) Both a and b are false. |

C



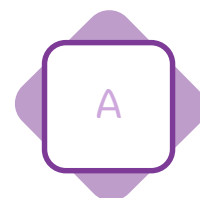
15) Choose the correct answer:

| | |
|--|--|
| A) Authorization occurs before authentication. | C) Authorization is visible to the user. |
| B) Authorization is the verification of the user's identity. | D) All of the above are incorrect. |



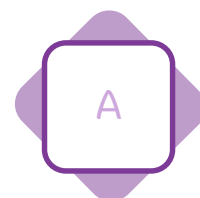
16) What does SOAP stand for?

| | |
|--|--|
| A) Simple Object Access Protocol. | C) Secure Online Application Protocol. |
| B) Structured Object Authentication Process. | D) Standardized Online Application Platform. |



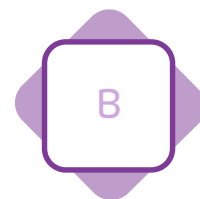
17) Choose the correct answer:

| | |
|---|---|
| A) Using an access token with a refresh token is more secure. | C) A refresh token can be used without an access token. |
| B) The lifespan of a refresh token is shorter than that of an access token. | D) All of the above are incorrect. |



18) What is the method that relies on sending a Token with each request in the Authentication process?

| | |
|--------------------------|---------------|
| A) Basic Authentication. | C) API Key. |
| B) Bearer Token. | D) OAuth 2.0. |





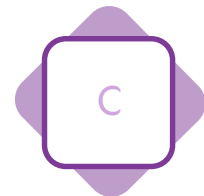
19) What is the method that relies on the device's IP address or MAC address in the Authentication process?

A) Basic Authentication.

C) API Key.

B) Bearer Token.

D) OAuth 2.0.



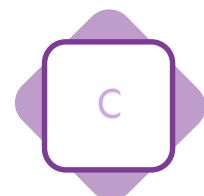
20) Which tool is used to test REST services?

A) WSDL

C) Postman

B) SOAP

D) XML



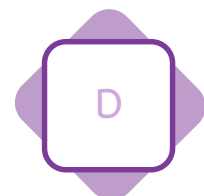
21) Choose the correct answer:

A) REST services support all types of formats and consume less bandwidth, while SOAP services support only XML and consume more bandwidth.

C) REST works only on HTTP/HTTPS protocols and is easier to implement, while SOAP is harder to implement and works on any protocol.

B) SOAP services have built-in security, while REST does not.

D) All of the above are correct.



"أنت أقوى مما تعتقد، والنجاح ينتظر أولئك
الذين يجرؤون على السعي وراء أحلامهم"



المحاضرة السادسة

1) What does RBAC (Role-Based Access Control) mean?

A) Giving permissions based on time

C) Giving permissions based on location

B) Giving permissions based on roles

D) Giving permissions based on devices

B

2) What is the best solution in RBAC if there are many accountants needing the same permissions?

A) Assign permissions to each user individually

C) Create a **Role** for accountants and give it permissions once

B) Assign permissions through email

D) Create a **Group** without permissions

C

3) What is the main difference between Role and Group in RBAC?

A) Group gives permissions to only one user

C) Role contains more than one Group

B) Group includes multiple Roles together

D) Group gives permissions to specific data only

B

4) Which type of authorization is based on Attributes instead of Roles?

A) ABAC

C) Group-Based Access

B) RBAC

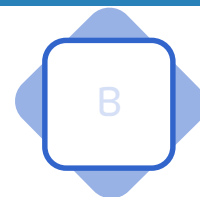
D) Static Access

A



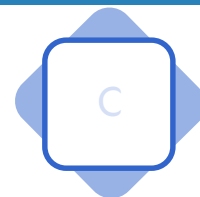
5) In ABAC, which level depends on user information like name or age?

| | |
|----------------|-------------|
| A) Environment | C) Resource |
| B) User | D) Role |



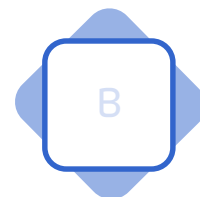
6) What is the main goal of using ABAC instead of RBAC?

| | |
|-----------------------------|------------------------------|
| A) Making the system faster | C) Providing higher security |
| B) Improving user interface | D) Reducing server costs |



7) What is an example of Environment level in ABAC?

| | |
|--|---|
| A) Giving permissions to a user based on ID | C) Giving permissions based on the requested resource |
| B) Allowing access to data only during certain hours | D) Assigning roles based on jobs |



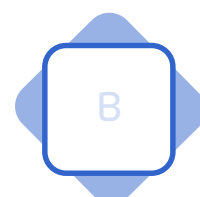
8) In ABAC, which level depends on the requested resource?

| | |
|---------|----------------|
| A) User | C) Environment |
| B) Role | D) Resource |



9) If you want to give both Managers and Accountants the same permissions, what is the best way in RBAC?

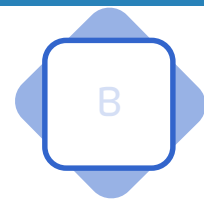
| | |
|---|---|
| A) Create a Role for each user | C) Assign permissions to each user manually |
| B) Create a Group that includes both Managers and Accountants | D) Use Attributes for each user |





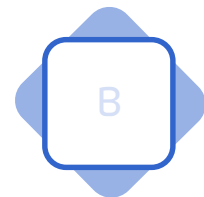
10) Which type of authorization is mostly used in high-security systems?

| | |
|---------|-----------------------|
| A) RBAC | C) Group-Based Access |
| B) ABAC | D) Static Access |



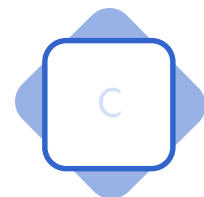
11) What is the main problem with storing data in regular files?

| | |
|---|-------------------------|
| A) Easy searching | C) Easy access |
| B) Data repetition and difficult modification | D) Saving storage space |



12) What is the main benefit of using databases?

| | |
|----------------------|--|
| A) Repeating data | C) Easy access and modification without repetition |
| B) Only storing data | D) Increasing data size |



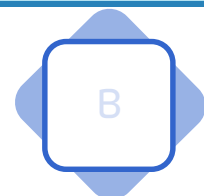
13) Which type of database uses tables and relationships?

| | |
|----------|-------------|
| A) SQL | C) Document |
| B) NoSQL | D) Graph |



14) What format is used in NoSQL Document databases?

| | |
|---------|---------|
| A) XML | C) CSV |
| B) JSON | D) HTML |





15) What is an example of a Document NoSQL database?

- | | |
|------------|-----------|
| A) MySQL | C) Oracle |
| B) MongoDB | D) SQLite |

B

16) Which type of database stores data in Key-Value pairs?

- | | |
|--------------|------------|
| A) Redis | C) MySQL |
| B) Cassandra | D) MongoDB |

A

17) Which database stores data as Nodes and Edges?

- | | |
|------------|-----------|
| A) MongoDB | C) Neo4j |
| B) Redis | D) SQLite |

C

18) Which type of NoSQL database uses dynamic columns?

- | | |
|----------------|--------------|
| A) SQL | C) Document |
| B) Wide Column | D) Key Value |

B

19) What is an example of a Wide Column database?

- | | |
|--------------|-----------|
| A) Cassandra | C) Redis |
| B) MongoDB | D) Oracle |

A

"لا تدع الخوف من الفشل يمنعك من المحاولة"
الفشل هو مجرد خطوة نحو النجاح

19) What is the main purpose of Shared Preferences in mobile applications?

- | | |
|--|-------------------------------|
| A) Storing images and videos | C) Storing data on the server |
| B) Storing app settings like language or theme | D) Storing text messages |

B

20) Which type of storage keeps data inside the device without the internet?

- | | |
|---------------------|-----------------------|
| A) Cloud Storage | C) Shared Preferences |
| B) Internal Storage | D) SQL |

B

21) What type of storage is used to save user preferences like Dark Mode or Language?

- | | |
|-----------------------|---------|
| A) Internal Storage | C) SQL |
| B) Shared Preferences | D) JSON |

B

22) Which database is used locally in mobile applications?

- | | |
|------------|--------------|
| A) MySQL | C) SQLite |
| B) MongoDB | D) Cassandra |

C

23) Which database is commonly used on servers?

- | | |
|-----------|-----------------------|
| A) MySQL | C) Shared Preferences |
| B) SQLite | D) Internal Storage |

A

24) Which type of database is best for large, unstructured data like social media posts?

- | | |
|-----------------------|---------------------|
| A) SQL | C) Internal Storage |
| B) Shared Preferences | D) NoSQL |

D



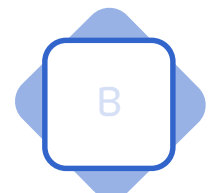
25) In ABAC, which level depends on the requested resource?

| | |
|---------|----------------|
| A) User | C) Environment |
| B) Role | D) Resource |



26) Which of the following statements best describes ABAC?

| | |
|---|--|
| A) It assigns permissions to users based on their roles. | C) It assigns permissions to users based on their groups. |
| B) It assigns permissions to users based on their attributes. | D) It assigns permissions to users based on their environment. |



27) In RBAC, what does assigning a "Role" achieve?

| | |
|--|---|
| A) It provides individual users with specific permissions. | C) It groups multiple roles into one. |
| B) It assigns permissions to an attribute. | D) It provides users with permissions based on their roles. |



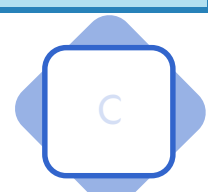
28) Which of the following is NOT a type of NoSQL database?

| | |
|--------------|----------------|
| A) Document | C) Wide Column |
| B) Key Value | D) SQL |



29) In a Graph database, what does an "Edge" represent?

| | |
|--------------------------------|----------------------------------|
| A) A node in the database. | C) A relationship between nodes. |
| B) A document in the database. | D) An attribute of a node. |



30) What is the main advantage of using a NoSQL database like MongoDB for social media applications?

A) It uses fixed schemas for data storage.

C) It restricts the number of records stored.

B) It handles large volumes of unstructured data.

D) It requires complex table relationships.

B

31) In ABAC, what does "Environment" level control?

A) Permissions based on user attributes.

C) Permissions based on environmental attributes like time and location.

B) Permissions based on the resource requested.

D) Permissions based on user roles.

C

32) What is the primary function of Internal Storage in mobile applications?

A) To store data on a remote server.

C) To manage user preferences.

B) To store data locally on the device.

D) To execute API calls.

B

33) Which type of database is most suitable for applications requiring dynamic schema adjustments?

A) SQL Database

C) Wide Column Database

B) Key Value Database

D) Graph Database

C

34) Which type of database is most suitable for applications requiring dynamic schema adjustments?

A) SQL Database

C) Wide Column Database

B) Key Value Database

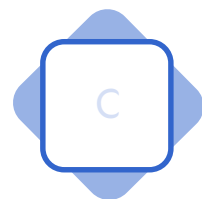
D) Graph Database

C



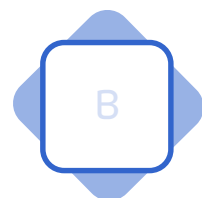
35) What is a significant disadvantage of storing data in files as opposed to databases?

- | | |
|---|---|
| A) Files have unlimited storage capacity. | C) Files often result in data duplication and difficulty in modification. |
| B) Files simplify data retrieval. | D) Files support complex query languages. |



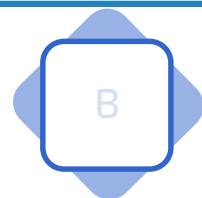
36) Which of the following best describes the purpose of a logging activity in databases?

- | | |
|--|-----------------------------------|
| A) To create backup files. | C) To reduce the database size. |
| B) To track and record changes made to the database. | D) To optimize query performance. |



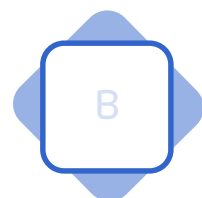
37) Which type of NoSQL database stores data as JSON-like documents?

- | | |
|--------------|----------------|
| A) Key Value | C) Wide Column |
| B) Document | D) Graph |



38) What is the main benefit of using SQLite for mobile applications?

- | | |
|--|---|
| A) It stores data on the server. | C) It requires internet connection for data access. |
| B) It provides a lightweight, local database for the device. | D) It stores user preferences. |



39) Which type of control grants permissions to users based on a group they belong to?

A) RBAC

C) MAC (Mandatory Access Control)

B) ABAC

D) DAC (Discretionary Access Control)

A

40) Which type of control grants permissions to users based on a group they belong to?

A) RBAC

C) MAC (Mandatory Access Control)

B) ABAC

D) DAC (Discretionary Access Control)

A

41) Which of the following is a disadvantage of using Internal Storage in mobile applications?

A) Fast access to data.

C) Potential data loss if the device is damaged or lost.

B) Requires internet connection.

D) Data remains consistent across devices.

C

42) In a Wide Column NoSQL database, what is the primary feature that differentiates it from SQL databases?

A) Fixed schema for tables.

C) Predefined table relationships.

B) Unlimited number of dynamic columns.

D) Data stored as JSON documents.

B

43) Which type of NoSQL database is best suited for analyzing complex relationships between entities?

A) Key Value

C) Graph

B) Document

D) Wide Column

C



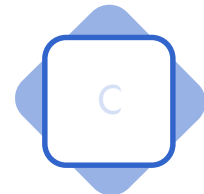
44) What is one of the main purposes of a database log?

A) Encrypting data.

C) Tracking database modifications for audit purposes.

B) Visualizing data relationships.

D) Storing backup copies of data.



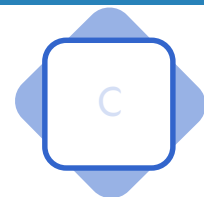
45) Which localization strategy involves using Google Translate API for translation?

A) Backend API handling

C) Dynamic in-app translation

B) Frontend processing

D) Server-side translation dictionary



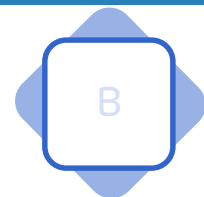
46) What type of database structure does Redis use?

A) Document

C) Wide Column

B) Key Value

D) Graph



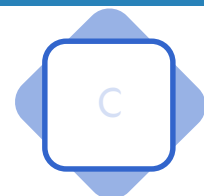
47) Which type of NoSQL database structure is used by Cassandra?

A) Document

C) Wide Column

B) Key Value

D) Graph



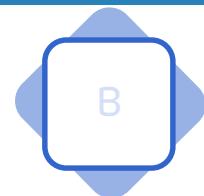
48) Which NoSQL database stores data in the form of key-value pairs?

A) MongoDB

C) Neo4j

B) Redis

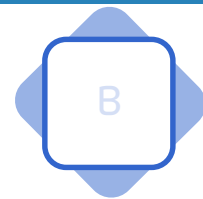
D) Cassandra





49) Which NoSQL database type is most appropriate for handling dynamic columns without a fixed schema?

| | |
|----------------|--------------|
| A) SQL | C) Document |
| B) Wide Column | D) Key Value |



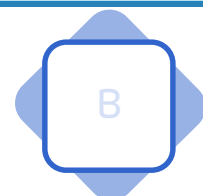
50) What type of NoSQL database is Neo4j known for?

| | |
|----------------|-------------|
| A) Key Value | C) Document |
| B) Wide Column | D) Graph |



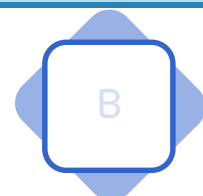
51) Which type of database would you choose for storing hierarchical data like organizational charts or social network connections?

| | |
|----------|--------------|
| A) SQL | C) Document |
| B) Graph | D) Key Value |



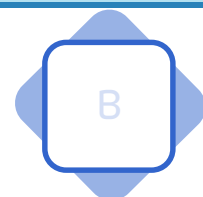
52) What type of database structure does MongoDB use?

| | |
|--------------|----------------|
| A) Key Value | C) Wide Column |
| B) Document | D) Graph |



54) Which of the following web services is considered flexible and more commonly used:

| | |
|---------|----------------------|
| A) SOAP | C) Both |
| B) REST | D) None of the above |





الاختصارات

1) What does URL stand for?

A) Uniform Resource Locator

C) Uniform Resource Link

B) Universal Resource Locator

D) Universal Resource Link

A

2) What does DNS stand for?

A) Domain Network Service

C) Domain Network System

B) Domain Name System

D) Domain Name Service

B

3) What does API stand for?

A) Application Programming Interface

C) Application Protocol Interface

B) Application Programming Language

D) Application Protocol Language

A

4) What does HTTP stand for?

A) Hypertext Transmission Protocol

C) Hypertext Transport Protocol

B) Hypertext Transfer Protocol

D) Hypertext Translation Protocol

B

5) What does ISP stand for?

A) Internet Service Provider

C) Internet System Provider

B) Internet Service Proxy

D) Internet System Proxy

A

6) What does NoSQL stand for?

| | |
|---------------------------------------|-----------------|
| A) Not Only Simple Query Language | C) Not Only SQL |
| B) Not Only Structured Query Language | D) B & C |

D

7) What does SQL stand for?

| | |
|------------------------------|----------------------------|
| A) Simple Query Language | C) System Query Language |
| B) Structured Query Language | D) Standard Query Language |

B

8) What does WSDL stand for?

| | |
|------------------------------------|-------------------------------------|
| A) Web Service Data Language | C) Web Service Description Language |
| B) Web Service Definition Language | D) Web Service Deployment Language |

C

9) What does CRUD stand for?

| | |
|-------------------------------------|--------------------------------------|
| A) Create, Retrieve, Update, Delete | C) Create, Read, Update, Develop |
| B) Create, Read, Update, Delete | D) Create, Retrieve, Update, Develop |

B

10) What does RBAC stand for?

| | |
|------------------------------|--------------------------------------|
| A) Role-Based Access Control | C) Role-Biased Access Control |
| B) Role-Bused Access Control | D) Role-Based Authentication Control |

A

11) What does ABAC stand for?

| | |
|-----------------------------------|---|
| A) Attribute-Based Access Control | C) Attribute-Biased Access Control |
| B) Attribute-Bused Access Control | D) Attribute-Based Authentication Control |

A

12) What does MVP stand for in product development?

| | |
|---------------------------|-------------------------------|
| A) Minimum Viable Product | C) Minimum Valuable Presenter |
| B) Most Valuable Product | D) Most Viable Product |

A

13) What does MVC stand for?

| | |
|--------------------------|---------------------------|
| A) Model View Content | C) Model View Component |
| B) Model View Controller | D) Model Value Controller |

B

14) What does MVP stand for?

| | |
|-------------------------|-----------------------------|
| A) Model View Presenter | C) Model Valuable Presenter |
| B) Model View Product | D) Model Value Product |

A

15) What does MVVM stand for?

| | |
|---------------------------|---------------------------|
| A) Model View View Model | C) Model View Value Model |
| B) Model Value View Model | D) Model View View Method |

A

16) What does SOAP stand for?

| | |
|---------------------------------------|--------------------------------------|
| A) Simple Object Access Protocol | C) Simple Object Access Program |
| B) Simple Object Application Protocol | D) Simple Object Application Program |

A

17) What does REST stand for?

| | |
|--|---|
| A) Representational State Transfer | C) Representational System Transfer |
| B) Representational State Transmission | D) Representational System Transmission |

A



القسم العملي

HTML

1. Which tag encloses the complete HTML document?

A) <head>

C) <html>

B) <body>

D) <title>

C

2. Which tag represents the document's header?

A) <head>

C) <html>

B) <body>

D) <title>

A

3. Which tag is used inside the <head> tag to mention the document title?

A) <title>

C) <meta>

B) <header>

D) <link>

A

4. Which tag represents the document's body?

A) <html>

C) <div>

B) <head>

D) <body>

D

5. Which tag represents a paragraph?

A) <h1>

C) <div>

B) <p>

D)

B

6. What is an HTML element defined by?

A) A starting tag

C) Both a starting and closing tag

B) A closing tag

D) None of the above

C

7. What does an HTML element contain if it has other content inside it?

A) A closing tag

C) Nested elements

B) An attribute

D) Comments

C

8. Which tag is used to create headings?

A) <h1>

C) <header>

B) <title>

D) <head>

A

9. Which tag is used to create a line break in HTML?

A) <hr>

C) <line>

B)

D) <separator>

B

10. What is the purpose of the tag in HTML?

A) To insert an image

C) To define a section

B) To create a hyperlink

D) To create a table

A

11. Which tag is used to create an unordered list?

A)

C)

B)

D) <list>

A

12. Which tag is used to create an ordered list?

A)

C)

B)

D) <list>

B

13. How can comments be added in HTML code?

A) <!-- This is a comment -->

C) // This is a comment

B) /** This is a comment **/

D) # This is a comment

A

14. What attribute is used to specify the URL of an image in the tag?

A) src

C) link

B) href

D) url

A

15. What attribute is used to specify the URL of an image in the tag?

A) src

C) link

B) href

D) url

A

16. Which tag is used to create a table in HTML?

A) <tr>

C) <th>

B) <td>

D) <table>

D

17. Which tag is used to define a table row in HTML?

A) <table>

C) <tr>

B) <td>

D) <th>

C

18. Which tag is used to define a table cell in HTML?

A) <th>

C) <td>

B) <tr>

D) <table>

C

19. Which attribute is used to set the background color of an HTML element?

A) bgcolor

C) style

B) background-color

D) color

C

(with the background-color property)

20. Which tag is used to define a hyperlink in HTML?

A) <link>

C) <href>

B) <a>

D) <url>

B

21. What is the purpose of the <meta> tag in the document's head?

A) To specify metadata

C) To create links

B) To define the title

D) To include styles

A

22. Which attribute is used to align text within an HTML element?

A) align

C) text-align

B) valign

D) style

A

23. Which attribute is used to specify the width of an element?

A) height

C) width

B) size

D) length

C



24. Which attribute is used to specify the height of an element?

A) height

C) width

B) size

D) length

A

25. Which tag is used to include an image in an HTML document?

A)

C) <picture>

B) <image>

D) <media>

A

26. How can you set the width and height of an image in HTML?

A) Using the width and height attributes in the tag

C) Using CSS

B) Using the style attribute

D) All of the above

D

27. Which tag is used to create a hyperlink to another document in HTML?

A) <link>

C) <href>

B) <a>

D) <url>

B

28. Which tag is used to define preformatted text in HTML?

A) <code>

C) <samp>

B) <pre>

D) <kbd>

B

29. Which tag is used to define a section in an HTML document?

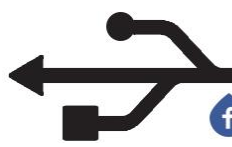
A) <div>

C) <article>

B) <section>

D) All of the above

D



CSS

1. What does CSS stand for?

A) Creative Style Sheets

C) Computer Style Sheets

B) Cascading Style Sheets

D) Colorful Style Sheets

B

2. Which property is used to change the background color of an element?

A) color

C) background-color

B) bgcolor

D) background

C

3. Which CSS property is used to change the text color of an element?

A) fgcolor

C) color

B) text-color

D) font-color

C

4. Which property is used to change the font of an element?

A) font-weight

C) font-style

B) font-family

D) font-variant

B

5. How do you make each word in a text start with a capital letter?

A) text-transform:capitalize

C) text-transform:lowercase

B) text-transform:uppercase

D) text-transform:initial

A

6. Which property is used to change the left margin of an element?

A) padding-left

C) indent

B) margin-left

D) spacing

B

7. How do you add a background color for all <h1> elements?

- | | |
|---------------------------------------|---------------------------------------|
| A) h1 {background-color:#FFFFFF;} | C) all.h1 {background-color:#FFFFFF;} |
| B) h1.all {background-color:#FFFFFF;} | D) h1 {bgcolor:#FFFFFF;} |

A

8. Which property is used to change the space between characters in text?

- | | |
|-------------------|----------------------|
| A) spacing | C) text-spacing |
| B) letter-spacing | D) character-spacing |

B

9. How do you make text bold in CSS?

- | | |
|----------------------|-------------------------|
| A) font-weight: bold | C) font-variant: bold |
| B) font-style: bold | D) font-thickness: bold |

A

10. Which property is used to add or change the border color of an element?

- | | |
|-----------------|-----------------|
| A) border-color | C) border-width |
| B) border-style | D) border-bg |

A

11. Which property is used to add or change the inner space of an element?

- | | |
|------------|------------|
| A) padding | C) spacing |
| B) margin | D) border |

A

12. How do you select an element with id "header"?

- | | |
|------------|------------|
| A) #header | C) header |
| B) .header | D) *header |

A

13. How do you select elements with class name "main"?

A) *main

C) #main

B) .main

D) main

B

14. Which property is used to change the width of an element?

A) height

C) size

B) width

D) length

B

15. Which property is used to change the height of an element?

A) height

C) size

B) width

D) length

A

16. Which property is used to align text?

A) align

C) text-align-center

B) text-align

D) font-align

B

1. How do you create a dotted border using CSS?

A) border-style: dotted

C) border-style: dashed

B) border-style: solid

D) border-style: dot

A

2. Which CSS property controls the text size?

A) font-style

C) font-size

B) text-size

D) text-style

C

3. How do you make the text italic in CSS?

A) font: italic

C) text-style: italic

B) font-style: italic

D) text-italic: true

B

4. Which property is used to change the bottom margin of an element?

A) padding-bottom

C) bottom-margin

B) margin-bottom

D) margin

B

5. How do you make the first letter of every word uppercase in CSS?

A) text-transform:capitalize

C) text-transform:lowercase

B) text-transform:uppercase

D) text-transform:initial

A

6. Which property is used to change the font size of an element?

A) font-size

C) font-style

B) text-size

D) text-style

A

7. How do you make the text alignment right in CSS?

A) text-align:right

C) text-right

B) align:right

D) align-text:right

A

8. Which property is used to set the color of a text?

A) color

C) fgcolor

B) text-color

D) font-color

A

9. How do you set the background image in CSS?

- | | |
|---------------------------------------|---------------------------------|
| A) background-image: url('image.jpg') | C) bgimage: url('image.jpg') |
| B) bg-image: url('image.jpg') | D) background: url('image.jpg') |

A

10. Which property is used to change the space between lines of text?

- | | |
|----------------|-----------------|
| A) spacing | C) text-height |
| B) line-height | D) line-spacing |

B

11. How do you set the display property of an element to be hidden?

- | | |
|-----------------------|---------------------|
| A) visibility: hidden | C) visibility: none |
| B) display: hidden | D) display: none |

D

12. Which property is used to change the style of the list marker?

- | | |
|-----------------|----------------|
| A) list-style | C) list-marker |
| B) marker-style | D) marker |

A

13. How do you select an element with the class name "button" and id "submit"?

- | | |
|-------------------|------------------|
| A) .button#submit | C) button#submit |
| B) #submit.button | D) submit.button |

A

14. Which CSS property controls the element's float behavior?

- | | |
|------------------|-------------|
| A) float | C) position |
| B) float-element | D) display |

A

JavaScript

1. What is JavaScript primarily used for?

A) Server-side scripting

C) Client-side scripting

B) Styling web pages

D) Database management

C

2. Which symbol is used for single-line comments in JavaScript?

A) //

C) <>

B) /*

D) #

A

3. How do you create a function in JavaScript?

A) function = myFunction() { }

C) function:myFunction() { }

B) function myFunction() { }

D) function::myFunction() { }

B

4. How do you call a function named "myFunction" in JavaScript?

A) call myFunction()

C) myFunction()

B) call function myFunction()

D) myFunction;

C

5. How do you write a conditional statement for executing some code if "i" is equal to 5?

A) if i = 5 then

C) if (i == 5)

B) if i == 5 then

D) if i = 5

C

6. How does a while loop start in JavaScript?

A) while (i <= 10 i++)

C) while i = 1 to 10

B) while (i <= 10; i++)

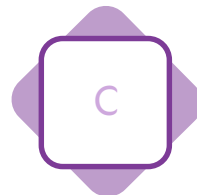
D) while (i <= 10)

D



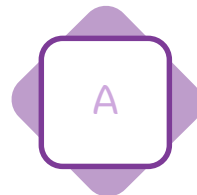
7. Which of the following is not a reserved word in JavaScript?

| | |
|--------------|------------|
| A) interface | C) program |
| B) throws | D) short |



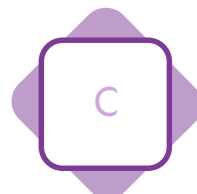
8. Which method is used to write HTML output in JavaScript?

| | |
|----------------------|---------------------|
| A) document.write() | C) document.print() |
| B) document.output() | D) document.show() |



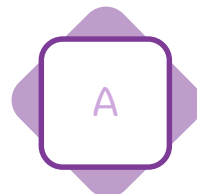
9. Which event occurs when the user clicks on an HTML element??

| | |
|-----------------|-------------|
| A) onmouseover | C) onclick |
| B) onmouseclick | D) onchange |



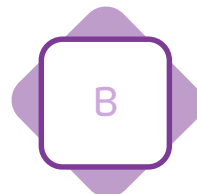
10. How do you declare a JavaScript variable?

| | |
|----------------------|--------------------|
| A) var carName; | C) v carName; |
| B) variable carName; | D) string carName; |



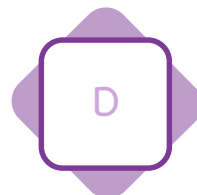
11. Which operator is used to compare two values in JavaScript?

| | |
|-------|--------|
| A) = | C) === |
| B) == | D) != |



12. Which method can be used to convert a string to a number in JavaScript?

| | |
|---------------|---------------------|
| A) parseInt() | C) parseFloat() |
| B) Number() | D) All of the above |



13. How do you add a property to an object in JavaScript?

A) obj["name"] = "John";

C) Both A and B

B) obj.name = "John";

D) None of the above

C

Laravel

1. Which command is used to create a new Laravel project using Composer?

A) php artisan new project-name

C) laravel new project-name

B) composer create-project --prefer-dist laravel/laravel project-name

D) php artisan create-project laravel/laravel project-name

B

2. To start the local development server in a Laravel project, which command should you use?

A) php artisan start

C) php artisan serve

B) php artisan run

D) php artisan launch

C

3. Which command is used to create a new controller in Laravel?

A) php artisan make:controller ControllerName

C) php artisan generate:controller ControllerName

B) php artisan create:controller ControllerName

D) php artisan new:controller ControllerName

A

4. To create a new model and its corresponding migration file in Laravel, which command would you use?

A) php artisan make:model ModelName

C) php artisan make:model ModelName -m

B) php artisan make:model ModelName -migration

D) Both B and C

D

Explanation: Both options B (--migration) and C (-m) will create a migration file along with the model.

5. What is the command to run all outstanding migrations in your Laravel application?

A) php artisan migrate

C) php artisan migration:run

B) php artisan db:migrate

D) php artisan migrate:up

A

6. Which command is used to roll back the last batch of migrations?

A) php artisan migrate:reset

C) php artisan migrate:refresh

B) php artisan migrate:rollback

D) php artisan migrate:down

B

7. To create a new middleware in a Laravel application, the command is

A) php artisan make:middleware
MiddlewareName

C) php artisan generate:middleware
MiddlewareName

B) php artisan create:middleware
MiddlewareName

D) php artisan new:middleware
MiddlewareName

A

8. Which command is used to create a new migration in Laravel?

A) php artisan make:migration
create_table_name

C) php artisan generate:migration
create_table_name

B) php artisan create:migration
create_table_name

D) php artisan migration:create
create_table_name

A

9. To seed your database using Laravel's seeders, which command would you run?

A) php artisan db:seed

C) php artisan make:seed

B) php artisan migrate:seed

D) php artisan seed:run

A

10. To generate a factory for a model in Laravel, which command do you use?

A) php artisan make:factory
FactoryName

C) php artisan make:factory
FactoryName --model=ModelName

B) php artisan create:factory
FactoryName

D) Both A and C

D

Explanation: Option A creates a basic factory; option C specifies the associated model.

11. What is the command to generate an event in Laravel?

| | |
|---|---------------------------------------|
| A) php artisan make:event EventName | C) php artisan create:event EventName |
| B) php artisan generate:event EventName | D) php artisan event:make EventName |

A

12. Which of the following is NOT a default directory in a new Laravel project?

| | |
|-----------------|--------|
| A) tests | C) app |
| B) node_modules | D) bin |

D

13. In Laravel, where are language localization files stored?

| | |
|-------------------|---------------------------|
| A) resources/lang | C) config/lang |
| B) app/lang | D) resources/localization |

A

14. What is the purpose of the vendor directory in a Laravel project?

| | |
|--|-------------------------------|
| A) Store application-specific classes | C) Store compiled views |
| B) Store third-party packages installed via Composer | D) Store cached configuration |

B

15. Which configuration file in Laravel is used to set application-level settings like timezone and locale?

| | |
|------------------------|----------------------|
| A) config/app.php | C) config/config.php |
| B) config/settings.php | D) config/system.php |

A

16. Where would you place custom validation rules in a Laravel application?

A) app/Rules

C) app/Http/Validators

B) app/Validators

D) app/Validation

A

17. Which file is used to define the application's exception handling behavior?

A) app/Exceptions/Handler.php

C) config/exceptions.php

B) app/Http/Middleware/VerifyCsrfToken.php

D) bootstrap/exception.php

A

18. Which component in Laravel handles the logic between Models and Views?

A) Middleware

C) Routes

B) Controllers

D) Service Providers

B

19. In Laravel, Facades provide a static interface to classes available in the application's service container. Which directory contains the classes that Facades reference?

A) app

C) resources

B) config

D) storage

A

20. In Laravel, which architectural pattern is primarily utilized?

A) Model-View-Presenter (MVP)

C) Model-View-ViewModel (MVVM)

B) Model-View-Controller (MVC)

D) Presenter-First Pattern

B

21. Which folder in a Laravel project contains the controllers?

A) app/Http/Controllers

C) app/Http

B) app/Controllers

D) resources/controllers

A

22. In Laravel, Service Providers are stored in which directory?

| | |
|------------------|------------------------|
| A) app/Providers | C) bootstrap/Providers |
| B) app/Services | D) config/providers |

A

23. Which of the following is responsible for handling database interactions in Laravel?

| | |
|----------------|---------------|
| A) Controllers | C) Views |
| B) Models | D) Middleware |

B

24. Where are Laravel's Blade template files typically stored?

| | |
|--------------------|------------------------|
| A) resources/views | C) resources/templates |
| B) app/Views | D) public/views |

A

25. Which directory contains route definition files in a Laravel application?

| | |
|---------------|---------------------|
| A) app/Routes | C) config/routes |
| B) routes | D) resources/routes |

B

26. In Laravel, Middleware is used for:

| | |
|---|-----------------------------------|
| A) Handling HTTP requests and responses | C) Defining application constants |
| B) Managing database migrations | D) Compiling assets |

A

27. What is the primary purpose of the routes/web.php file in a Laravel application?

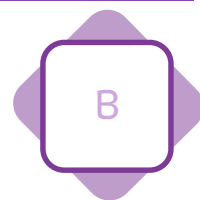
| | |
|---|-----------------------------------|
| A) Define API routes | C) Configure database connections |
| B) Define web routes accessible via a browser | D) Register service providers |

B



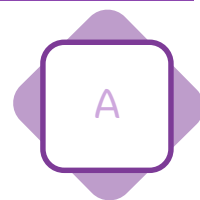
28. Which layer of the MVC pattern in Laravel is responsible for the presentation logic?

| | |
|----------|---------------|
| A) Model | C) Controller |
| B) View | D) Middleware |



29. The .env file in a Laravel project is used for:

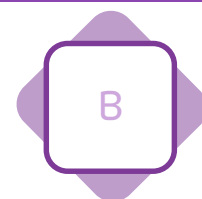
| | |
|--|----------------------------------|
| A) Defining environment-specific configuration | C) Storing Blade templates |
| B) Defining routes | D) Managing package dependencies |



Flutter

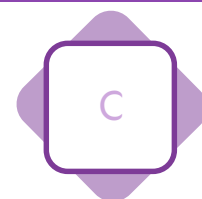
1. Which language is used to code Flutter apps?

| | |
|---------|-----------|
| A) Java | C) Kotlin |
| B) Dart | D) Swift |



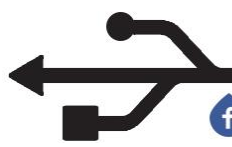
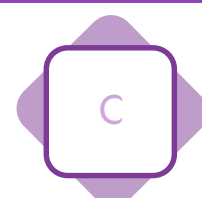
2. Which widget is used for layouts that need to be scrollable?

| | |
|--------------|--------------------------|
| A) Container | C) SingleChildScrollView |
| B) Column | D) Stack |



3. How do you add an image asset to a Flutter project?

| | |
|---|---|
| A) Place the image in the lib folder | C) Add the image path to the assets section in pubspec.yaml |
| B) Reference the image URL directly in code | D) Use the ImagePicker plugin |



4. What is the purpose of the pubspec.yaml file?

A) To define the app's routes

C) To configure database connections

B) To list the project's dependencies and assets

D) To set up user authentication protocols

B

5. Which widget would you use to create a list of items from a large data source efficiently?

A) ListView

C) ListView.builder

B) Column

D) GridView

C

6. What does the setState() method do in a StatefulWidget?

A) Initializes the state

C) Disposes of resources

B) Rebuilds the widget with updated values

D) Handles user inputs

B

7. Which widget allows you to specify how much space a child can occupy relative to its siblings?

A) Padding

C) SizedBox

B) Expanded

D) Align

B

8. How do you handle asynchronous operations in Dart?

A) Using threads

C) By blocking the main thread

B) Using the async and await keywords

D) Dart doesn't support asynchronous operations

B

9. Which widget allows for stacking widgets on top of each other?

A) Stack

C) Column

B) Row

D) ListView

A



10. How do you make a widget respond to taps in Flutter?

A) Wrap it with a GestureDetector

C) Use the TapHandler widget

B) Implement the onTap callback directly in the widget

D) Flutter doesn't support tap gestures

A

11. How do you handle asynchronous operations in Dart?

A) Using threads

C) By blocking the main thread

B) Using the async and await keywords

D) Dart doesn't support asynchronous operations

B

12. How do you apply padding to a widget?

A) By wrapping it in a Padding widget

C) By adjusting the margin property

B) By setting the padding property directly

D) Flutter doesn't support padding

A

13. What does the Expanded widget do in a Row or Column?

A) Shrinks its child to the smallest size possible

C) Centers its child

B) Forces its child to fill the available space

D) Makes its child scrollable

B

14. Which widget would you use to display a horizontal list in Flutter?

A) Row

C) Column

B) ListView with scrollDirection: Axis.horizontal

D) HorizontalList

B

15. What is the main role of the Navigator class?

A) Manages a stack of routes to move between different screens

C) Manages app settings

B) Handles network requests

D) Controls animations

A

16. Which widget is used to create a drop-down list in Flutter?

A) DropdownButton

C) ComboBox

B) PopupMenuButton

D) SelectMenu

A

17. What is the difference between final and const in Dart?

A) final variables can be set only once; const variables are compile-time constants

C) const variables can change; final variables cannot

B) They are the same

D) final is for functions; const is for variables

A

18. Which widget would you use to display a grid of images?

A) GridView

C) Table

B) ListView

D) Row

A

19. Which widget would you use to display a grid of images?

A) GridView

C) Table

B) ListView

D) Row

A

العالم بأسره يتنحى جانبا لِيُفسح الطريق
للإنسان الذي يعرف تماماً إلى أين يتوجه
GOOD LUCK ♥

